

Malhar Gudekar

(217) 766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

EDUCATION

University of Illinois Urbana-Champaign

M.S., Information Management

Aug 2024 – May 2026

GPA: 3.8/4.0

Coursework: Web Programming, Machine Learning & Cloud, Information Management, Methods of Data Science

University of Mumbai

B.E., Computer Science

Jun 2019 – May 2023

GPA: 9.25/10

Coursework: Data Structures & Algorithms, Machine Learning, NLP, Operating Systems, OOP

SKILLS

Languages: Python, SQL, Java, C++

Data Engineering & MDM: ETL/ELT Pipelines, Data Modeling, Semantic Layers, Batch and Streaming Pipelines, Data Quality Rules, Snowflake, PySpark, Airflow, AWS Glue, Step Functions

Analytics & Reporting: Power BI, Dashboards and Scorecards, KPI Monitoring, Self-Serve Analytics, Automated Reporting

Cloud & DevOps: AWS (S3, EC2, EKS), PostgreSQL, Neo4j, Git, CI/CD Pipelines, Docker, Automated Testing

WORK EXPERIENCE

University of Illinois Urbana-Champaign

May 2025 – Present

Research Assistant – iSchool

Champaign, IL

- Developed and maintained **AWS-based scalable data pipelines** to ingest high-volume wearable data for 100+ users, implementing **ETL workflows** with data quality checks, validation rules, and monitoring to ensure reliable downstream analytics
- Designed and optimized **SQL-based data models and warehouse schemas** in PostgreSQL, applying indexing, query tuning, and data modeling best practices, improving analytical query performance by **38%** across large datasets
- Implemented **workflow orchestration using AWS Step Functions** for real-time and batch pipelines; integrated logging, retries, and observability signals to support consistent data availability and platform reliability

Research Assistant – CHI AI in Infodemic Management

Jan 2025 – May 2025

- Designed and deployed **scalable ELT pipelines** in **Python** for high-volume social media datasets, reducing processing latency by **41%**; applied **automated testing** and CI/CD practices to ensure data integrity across pipeline stages
- Built distributed real-time ingestion pipelines using **Kafka** and **Neo4j**, enabling streaming data transformation and graph-based analytics to support unified views of large-scale datasets
- Identified and resolved data quality issues using targeted validation rules, process controls, and analytical reporting on pipeline health and data consistency

Business Intelligence Group

Aug 2025 – Dec 2025

Technical Consultant

Champaign, IL

- Built and deployed **scalable ETL/ELT data pipelines** using **Python**, FastAPI, and PostgreSQL to ingest, transform, and structure high-volume datasets for analytics and business decision-making
- Developed **dashboards and KPI monitoring systems** to define and track product and operational metrics, enabling teams to understand performance and usage trends
- Implemented **CI/CD pipelines** and automated testing frameworks to validate data pipelines and ensure reliability, consistency, and maintainability across environments

Swift Mobil Software Solutions Provider

Jul 2023 – Oct 2023

Data Analyst

Mumbai, IN

- Built **data pipelines feeding Power BI dashboards** for logistics KPIs and scorecards, reducing reporting time by **40%** and improving visibility into operational performance
- Processed large-scale datasets using **Python** and **PySpark distributed processing**, implementing transformations aligned with performance monitoring and analytics requirements
- Developed complex **SQL queries** to validate, reconcile, and analyze operational data, improving data quality and supporting faster root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

Jan 2023 – Jun 2023

Data Analyst

Mumbai, IN

- Built and maintained **SQL-based data pipelines supporting Power BI dashboards**, designing warehouse models and delivering analytics-ready datasets with enforced **data quality rules**
- Automated **data transformation workflows** using VBA with validation checks, error handling, and scheduling logic, improving processing efficiency and reducing reporting time by **42%**

Stu/Dio at Illinois

Aug 2025 – Present

Project Manager

Champaign, IL

- Led cross-functional team of 6 engineers delivering data-driven improvements and system tracking; managed **Agile workflows** and ensured timely delivery of analytics and platform features

PROJECTS

Scalable ML System for Customer Risk Prediction

Mar 2025 | UIUC Hackathon

- Built an **end-to-end ETL data pipeline** using Python, Pandas, and LightGBM to ingest, transform, and enrich large-scale time-series transaction data—incorporating **data quality checks** and feature engineering for model integration
- Designed **scalable feature engineering pipelines** capturing behavioral trends and temporal patterns, producing consistent and accurate analysis of key risk metrics and performance indicators
- Developed a **data-driven decision framework** with automated risk thresholds and pipeline outputs, improving prediction precision by **20%** and enabling scalable analytics workflows with low-touch reporting